1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In NMIBC treated with cisplatin-based neoadjuvant chemotherapy, a higher pretreatment tumor-stroma ratio (TSR) measured on H&E sections is causally associated with reduced therapeutic efficacy, due to ECM/stromal density creating diffusion barriers and stromal fibroblasts sequestering cisplatin, resulting in lower intratumoral cisplatin exposure and diminished tumor cell kill. This effect is most evident when TSR is assessed at the invasive front and is modulated by additional morphological features such as tumor budding and lymphocyte density, which reflect microenvironmental dynamics and invasion. The hypothesis predicts: (i) higher pretreatment TSR will be associated with lower rates of favorable therapy response (e.g., downstaging or pathologic complete response) after standardized cisplatin-based neoadjuvant therapy, independent of clinical covariates; (ii) ex vivo models with higher ECM density or fibroblast-rich stroma will show reduced cisplatin penetration and greater chemoresistance; (iii) quantitative shifts in TSR coupled with the proposed modifiers will improve the predictive performance beyond TSR alone in independent cohorts. No molecular/genomic markers or multi-gene signatures are required for this imaging-centered mechanism.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	tumor budding / single-cell invasion at invasive front	Supporting modifier from knowledge graph
Secondary	lymphocyte density (intratumoral and peritumoral)	Supporting modifier from knowledge graph
Secondary	spatial heterogeneity metrics (entropy of tumor/stroma distribution)	Supporting modifier from knowledge graph

Secondary	tumor region definition / invasive-front-focused TSR	Supporting modifier from knowledge graph
Secondary	segmentation reliability / scanner-stain calibration	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective and prospective digital pathology analyses to test the TSR–response association,

Prediction 2: validation across independent cohorts with standardized TSR measurement (invasive-front emphasis), and

Prediction 3: mechanistic ex vivo/in vitro diffusion studies using ECM-rich and fibroblast-containing models to demonstrate reduced cisplatin diffusion and efficacy under high stromal conditions. This plan intentionally excludes molecular markers to maintain a morphology-centric, mechanistic framework and to avoid overfitting in single-path models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- tumor budding / single-cell invasion at invasive front: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- lymphocyte density (intratumoral and peritumoral): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.